UNIT QUESTIONS

SKILLS CRITICAL THINKING 3

1

a Which of these objects is made from a magnetic material?

A a rubber band

B a piece of paper

C a steel nail

D a plastic ruler **(1)**

4

b In which of these situations will you see attraction?

A The north poles of two magnets are placed close together.

B A wooden rod is placed close to a strong magnet.

C The south poles of two magnets are placed close together.

D The north pole of one magnet is placed close to the south pole of another magnet. **(1)**

c Which of these statements is true for a magnetically hard material?

A They are difficult to bend.

B They are used to make permanent magnets.

C They are electrical insulators.

D They are used in electromagnets. **(1)**

(Total for Question 1 = 3 marks)

7

2

Copy and complete the following passage, filling in the blanks.

A wire carrying an electric current experiences a _____ when it is in a _____ field. This effect can be used in _____ and electric _____.

If a wire is moved through a magnetic field, or the magnetic field near a _____ of wire changes, a current can be _____ in the wire. This effect is used in _____ and _____.

(Total for Question 2 = 8 marks)

SKILLS REASONING

3

a Explain how a bicycle dynamo generates current. **(3)**

b Why does the dynamo produce no current when the cyclist has stopped? **(1)**

SKILLS CRITICAL THINKING 6

c What is an alternator? **(1)**

SKILLS INTERPRETATION, CRITICAL THINKING

d Explain using diagrams the difference between a.c. and d.c. currents. **(3)**

SKILLS CRITICAL THINKING 5

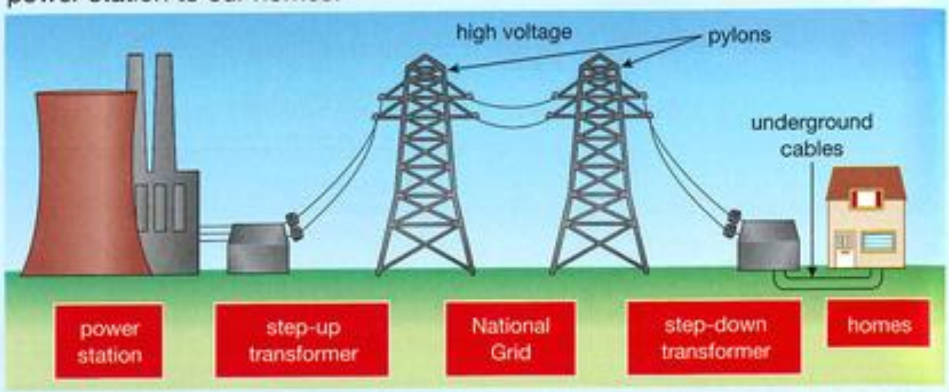
e Explain what is meant by this statement: 'The mains supply in the UK has a frequency of 50 Hz.' **(2)**

(Total for Question 3 = 10 marks)

PHYSICS ONLY

SKILLS REASONING

4 The diagram below shows how the electrical energy is transmitted from a power station to our homes.



7 **a** Why is the voltage of the supply increased before transmission? **(1)**

b Why is the voltage of the supply decreased before entering our homes? **(1)**

SKILLS PROBLEM SOLVING

8 **c** A step-up transformer has 100 turns on its primary coil and 20 000 turns on its secondary coil. Calculate the output voltage of the transformer if the input voltage is 12 V a.c. **(3)**

9 **d** Assuming that the transformer is 100% efficient, calculate the size of the induced current in the secondary coil if the current in the primary coil is 10 A. **(3)**

(Total for Question 4 = 8 marks)

END OF PHYSICS ONLY

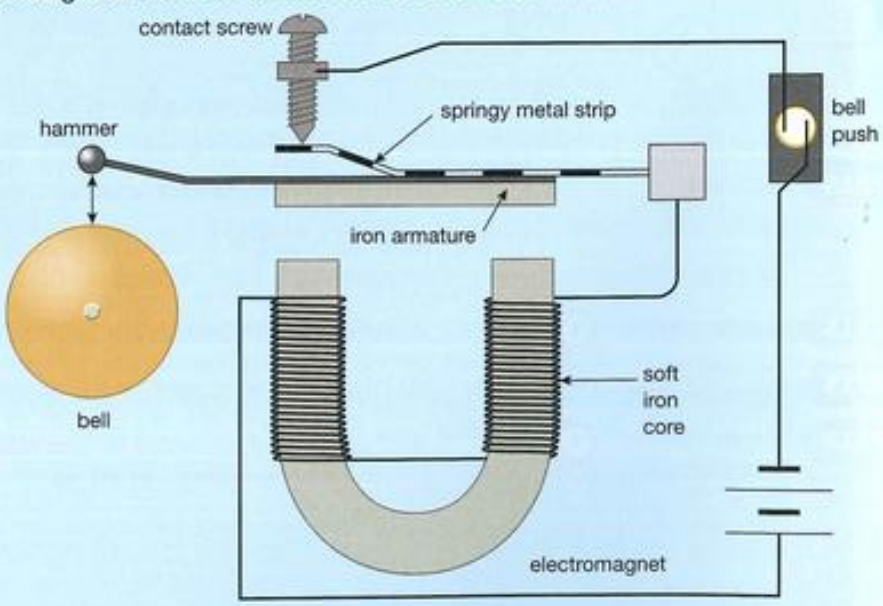
SKILLS CREATIVITY, REASONING

8 **5** A keen cyclist builds his own dynamo to generate electricity for his lights. Unfortunately, when he tries it out the lights are not bright enough. Suggest two changes he could make to his dynamo to generate more electricity. Is there a third solution to his problem?

(Total for Question 5 = 3 marks)

SKILLS REASONING

7 **6** The diagram below shows the circuit for an electric bell.



a Explain in your own words why the bell will not work if the electromagnet is replaced with a permanent magnet. **(2)**

b Explain why the core of the electromagnet used in an electric bell must not be made from steel. **(3)**

(Total for Question 6 = 5 marks)

PHYSICS ONLY

SKILLS REASONING 8

SKILLS PROBLEM SOLVING 9

7 **a** Explain the difference in the structure and use of a step-up transformer and a step-down transformer. **(4)**

b A transformer has 2000 turns on its primary coil and 6000 turns on its secondary coil. If an a.c. voltage of 12 V is applied across the primary coil calculate the voltage induced across the secondary coil. Assume that the transformer is 100% efficient. **(4)**

(Total for Question 7 = 8 marks)

END OF PHYSICS ONLY